

“Exams are not just a test of knowledge, but a tool for learning. Challenge yourself, identify where you struggle, and focus on those areas—learn and grow from your mistakes”

MCQ:**20x1=20**

1. return type of Math.round() is

- a) float b) double
- c) int d) char

2. System.out.println(Math.pow(2, Math.pow(49, 1/2)))

- a) 128 b) 128.0
- c) 1.0 d) 2.0

3. String s= “Hello”; int a=10; double b=20.0;

System.out.println(s+a+b);

- a) Hello1020.0 b) Hello1020
- c) Hello30.0 d) Hello10.020.0

4) State the type of loop:

for(int i=1; i != 6; i+=2) System.out.println(i);

- a) finite b) infinite
- c) fixed d) NULL

5) Absence of break causes fall through in

- a) if-else construct b) for loop
- c) switch-case d) do-while

6) Which one is in correct order of precedence

- a) ++ ! / - > b) ! / ++ - >
- c) / - ++ ! > d) / + ! ++ -

7. even if the condition is initially false _____ loop executes at least once

- 1. for 2. do-while
- 3. while 4. for-each

a) ?:

b) ()

c) {}

d) []

a) \n and \b b) \n and \c
c) \n and \t d) \n and \r

a) 0 to ≤ 10
b) 0 to < 10
b) 1 to ≤ 10
d) 1 to < 10

a) char, short, int, double b) short, char, int, double
c) double, int, short, char d) int, char, short, double

a) Implicit Typecasting/ Coercion b) Explicit
c) Automatic d) Error

a) Syntax Error b) Logical Error
b) Runtime Error/ Exception d) Compilation Error

a) 8 b) 32
c) 64 d) 16

a) .class and .java
b) .class and .java
c) .source and .java
c) .java and .byte

16) Give output of the following:

```
int k;  
for(k=5; k<=20; k+=7)  
    if(k%6==0) continue;  
System.out.println(k);
```

- a) Error
- b) 5
- b) 26
- d) 19

17. find output if n=10;

```
switch(n){  
    case 10: System.out.print(n*2);  
    case 4: System.out.print(n*4);  
        break;  
    default: System.out.print(n);  
}
```

- a) 2040
- b) 20
- b) 10
- d) Nothing

18) Ternary is a _____ operator

- a) Relational
- b) Conditional
- c) Arithmetic
- c) Logical

19. Which access specifier is used to restrict access of a member to the same package ?

- a) private
- b) public
- b) default
- d) protected

20. find output:

```
int i;  
for(i=5; i>10; i++) System.out.print(i); System.out.print(i*4);
```

- a) NO Output
- b) 5
- b) 20
- d) 5 6 7 8 9 10 44

1.

```
char ch= 'A';
for( int i=1; i<=4; i++){
    for(int j=1; j<=i; j++){

        System.out.print(ch);
        term +=2;
    }
    System.out.println();
}
```
2.

```
for(int i=1; i<=5; i++){
    for(int j=1; j<=5; j++){
        if(i==j || i+j==6 || i==1 || i==5 || j==1 || j==5){
            System.out.print("*");
        }
    }
    System.out.println();
}
```

Draw the star pattern generated this loop.

3. Write the following expression in JAVA.

$$k = \sqrt{ax + a^x + x^a}$$

4.

```
int x= 5, y=10;
x -= ++x + y-- + --y + x-- + x + 4%8;
x=? y=?
```

5. Find output and the number of times the loop executes

```
int n=4279;
while(n>0){
    int d= n%10;
    System.out.println(d);
    n/=100;
}
```

6. find output and the number times the loop executes

```
int p=1;
do{
    p*=2;
}while(p!=32);
```

7.

```
int a,b;
```

```
for( a=6, b=4; a<=24; a=a+6){
    if(a%b==0)break;
}
```

```
System.out.println(a + "\n" + b);
```

8. int a; float f; char ch; short s;

Assume an arithmetic expression is formed using the above variables and the result is stored in k. find data type of k.

9. Find the errors, write the type of errors you can see, and correct them (if possible)

```
class apple juice{
    public static void main{
        int a= 6%3;
        int b= a/a;
        System.out.println("Sum of two numbers="+b);
    }
}
```

```
10. for (int i = 1; i <= 3; i++) {
    int j = 1;
    while (j <= 2) {
        System.out.print(i + " " + j + " ");
        j++;
    }
}
```

Find output and number of times the loop executes.

Section B

Write programs in



and write VDTs

Solve any four:

15x4=60

1. Write a program to take distance and AC information (1 if AC is on and 0 if AC is off) from user calculate taxi fare according to tariff given below.

Distance	Fare per kilometre
upto 5 km	Rs. 150
Next 20 km	Rs. 22.50 per kilometre
Above 25 km	Rs. 30 per kilometre

The taxi charges Rs. 100 extra if the passenger wants the AC to be turned on.

2. Write a menu driven program to do the following:

(a) Print the number pattern using nested loop:

```
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1
```

(b) Print the following number pyramid using nested loop:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

3. Write a **menu driven** program to print Lucas Sequence or Pell Sequence based on users choice.

The **Lucas series** is a sequence of numbers similar to the Fibonacci sequence. Each term is the sum of the two preceding ones, starting with 2 and 1. The sequence follows the recurrence relation:

$$L(n)=L(n-1)+L(n-2) \text{ or } (a, b, a+b \dots)$$

1) Lucas Sequence: 2,1,3,4,7,11,18,29,47,76,... till N

The **Pell series** is a sequence of numbers defined by the recurrence relation:

$$P(n)=2\times P(n-1)+P(n-2) \text{ or } (a, b, a+2b \dots\dots\dots)$$

2) Pell Sequence: 0,1,2,5,12,29,70,169,408,985,...till N

4. Take a number and check if it's a duck number or not.

Duck number is a number such that it must contain 0 (one or more) but should not start with 0

5. Write a program to take x and n from user and find S.

$$S=\frac{x^0}{1!}+\frac{x^2}{3!}+\frac{x^4}{5!}+\frac{x^6}{7!}+\frac{x^8}{9!}+\dots\dots\dots+\frac{x^n}{(n+1)!}$$

6. Write a program to take N numbers from user. find the largest, smallest and average of all numbers.

